Design Proposal for Mach-Zehnder Interferometer Based Circuits

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***Index Terms*— Mach-Zehnder interferometer, silicon photonics.**

I. INTRODUCTION

Several PIC designs are proposed and fabricated in this document. The designs are based on Mach-Zehnder Interferometer (MZI) and are intended to be used to calculate the group index of a silicon on insulator (SOI) waveguide. The fabrication was done on a 220 SOI process. We compare simulated group index and measured group index and show that the measured group index is within the boundary of the corner analysis of our fabrication process. Lastly, a MZI based lattice filters are also proposed intended to be used to filter a wideband signal.

II. THEORY

A. Mach-Zehnder Interferometer

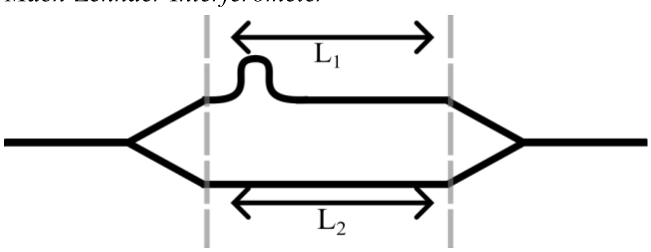

Fig. 1. Structure of an unbalanced MZI

Fig. 1 show a structure of an unbalanced MZI consisting of input y-branch, upper arm, lower arm and an output y-branch. It can be shown that at the output field of the MZI is

$$E_{\mathrm{o}} = \frac{E_i}{2}\left(\exp(-j\beta L_1) + \exp(-j\beta L_2)\right), \qquad (1)$$

where $E_{\mathrm{i,o}}$ is the input and output field respectively, $\beta = \frac{2\pi n_{\mathrm{eff}}}{\lambda}$ is the propagation constant of the waveguide. The intensity of the output field $|E_{\mathrm{o}}|^2$ is then can be written as

$$I_{\mathrm{o}} = \frac{I_{\mathrm{i}}}{2}[1 + \cos(\beta \Delta L)], \qquad (2)$$

where ΔL is the path length difference between the upper and lower arm.

B. MZI based lattice filter

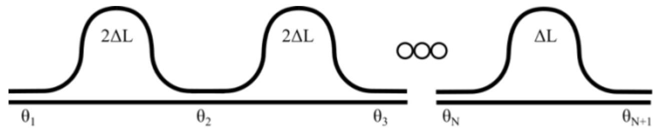

Fig. 2. General structure of optical lattice filter

Following [1], a lattice finite input response (FIR) filter can be implemented using power splitter and delay lines. Fig. 2 show a general implementation of optical based lattice filter. θ is the splitting ratio. To realize such filter in a PIC, the filter is implemented using a cascaded MZI where we use a directional coupler instead of a y-branches.

III. MODELING AND SIMULATION

A. Waveguide

In this design, we use strip waveguide of 220nm in height and 500nm in width. Fig. 3 and Fig. 4 show the electric field for the TE-mode and TM-mode of the waveguide at 1550nm. For the rest of the discussion and design, we only consider the TE-mode of the waveguide. Fig. 5 and Fig. 6 show the effective index and group index of the waveguide as a function of wavelength. It can be seen from the figures that the effective index of the waveguide decreases as a function of wavelength while the group index increases as a function of wavelength.

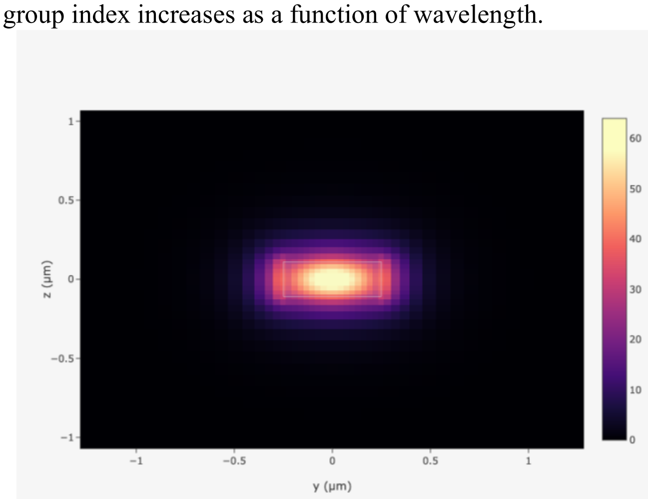

Fig. 3. Magnitude of electric field for TE mode

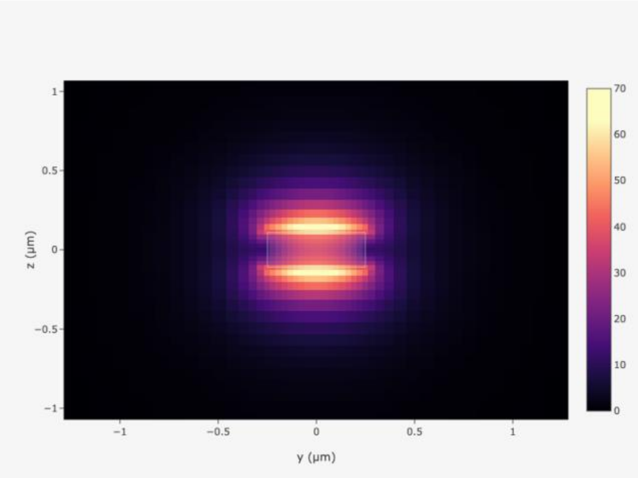

Fig. 4. Magnitude of electric field for TM mode

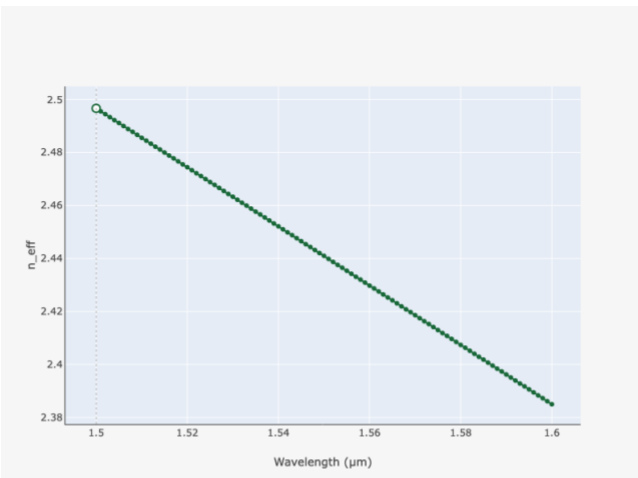

Fig. 5. Effective index of the waveguide as a function of wavelength

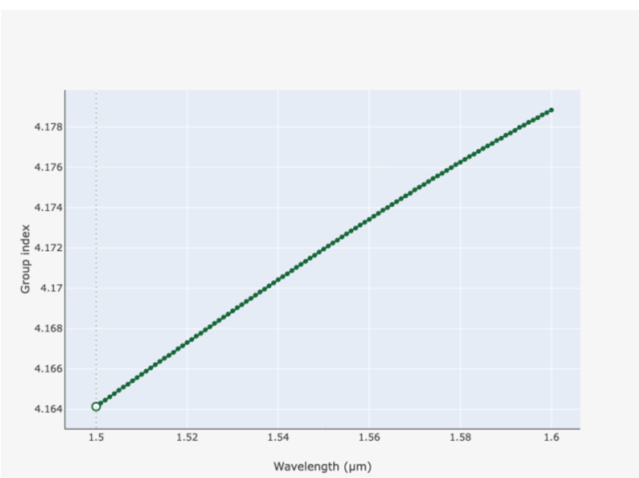

Fig. 6. Group index of the waveguide as a function of wavelength

From Fig. 5, a compact model of the waveguide is fitted to be

$$n_{\mathrm{eff}}(\lambda) = 2.441 - 1.116(\lambda - \lambda_0) - 0.05(\lambda - \lambda_0)^2, \quad (3)$$

where λ_0 is chosen to be 1.55 μm.

B. Mach-Zehnder Interferometer

Using the waveguide model, we simulate the transfer function of an MZI with ΔL of 50nm and 100nm. Fig. 7 shows that the simulation result. The results show that periodicity of the transfer functions follows ΔL as expected from equation (2).

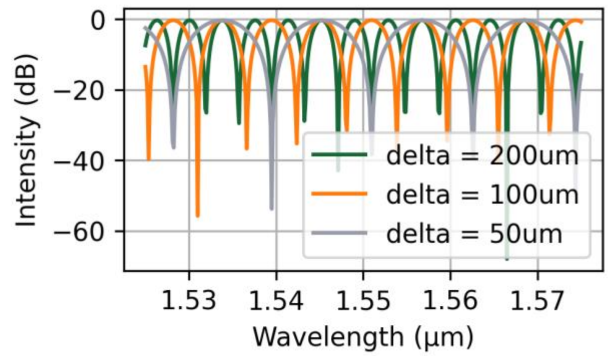

Fig. 7. Transfer function of MZIs

Given that there will be a pair of grating couplers in our circuit, Fig. 8 show the simulated end-to-end transfer function of our MZI where the envelop of the transfer function of the input and output grating coupler can be clearly seen.

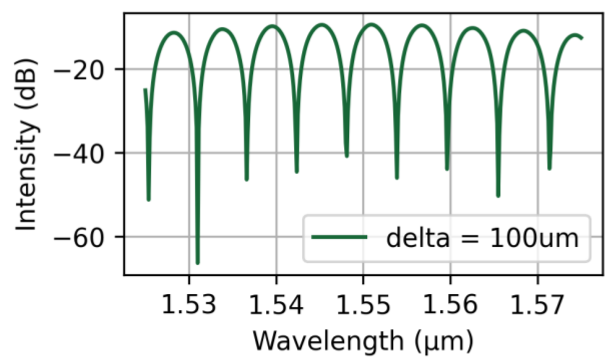

Fig. 8. End-to-end transfer function of an MZI with ΔL of 100 um

Further, from the transfer function of the MZI we can write the free spectral range of the MZI as

$$\mathrm{FSR} = \frac{\lambda^2}{n_{\mathrm{g}} \Delta L}, \quad (4)$$

where n_{g} is the group index of the waveguide. For a 100μm path difference, the FSR is around 5.77nm. In our design, we proposed 5 different MZI variations each with different path length. Table 1 shows the proposed ΔL and its corresponding FSR. Lastly, we add a loopback element to calibrate the insertion loss of the grating couplers.

Table 1 Proposed design variations of MZI

$\Delta L[\mu m]$	FSR[nm]
50	11.54
100	5.77
150	3.83
200	2.88
250	2.3

C. MZI based lattice filter

We limit ourselves to a 2 stages lattice filter. From [1] the splitting ratio for each of the power splitter are 6.69%, 75%, 50%. For the simulation, we already consider a MZI implementation of the power splitter. Using (2) and following [2], the corresponding ΔL of our MZI are 264.7nm, 105.87nm and 158.8nm respectively. We calculate the ΔL_f of our lattice filter from the FSR of the filter. We consider 2 FSRs of 12nm, and 6nm which correspond to ΔL_f of 48μm, and 96μm. Fig. 9 shows the simulated transfer function of the lattice filter.

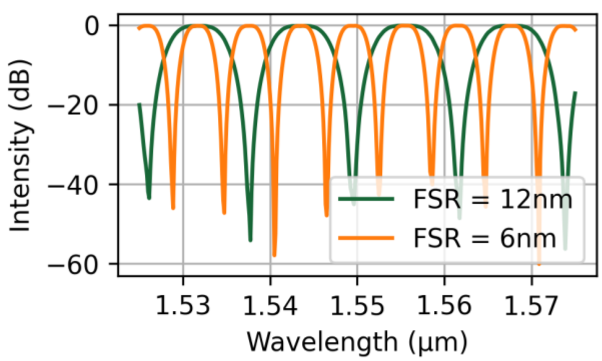

Fig. 9. Transfer function of 2 lattice filters

IV. Results and Discussions

The proposed devices were fabricated in a 220nm SOI process under a multi project wafer run. The resulting die were then measured in a photonic lab in UBC. Fig. 10 show the response of different MZI structures where the MZI transfer function can clearly be seen. From the measured data, we proceed with the calculation of FSR to obtain the group index of the silicon waveguide. For this calculation we limit ourselves to the 3dB bandwidth of the fabricated grating coupler which is between 1540nm and 1565nm. Table 2 shows the calculated FSR and the calculated group index at 1550nm of the silicon waveguide for different ΔLs of the MZI.

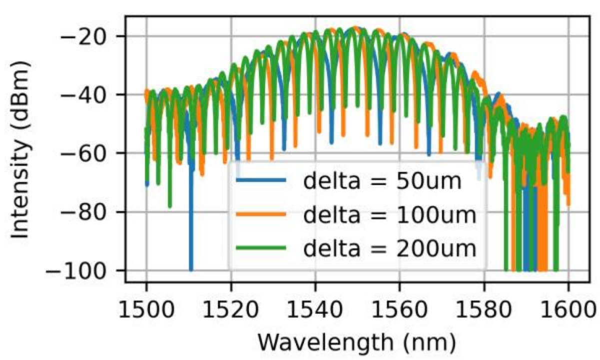

Fig. 10. Measured transfer function of the fabricated MZIs.

Table 2 Calculated FSR and group index at 1550nm as a function of ΔL

$\Delta L[\mu\text{m}]$	FSR[nm]	n_g
50	11.5	4.18
100	5.74	4.18
150	3.82	4.19
200	2.87	4.19
250	2.3	4.19

The table shows that the group index at 1550nm is around 4.19. This number lies within the range of the group index of 4.15 and 4.23 that we calculated using corner analysis.

Fig. *11* shows the transfer function of the fabricated MZI based lattice filter where the calculated FSR is 11.85nm and 6.02nm respectively. Although the FSR is close to the designed value the pass-band frequency response of the filter is not the same as the simulated design. This discrepancy could come from the phase response of the directional couplers or path length mismatch of the Mach-Zehnder arms.

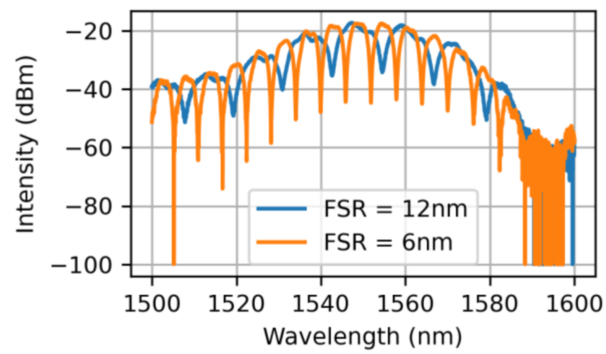

Fig. 11. Measured transfer function of the fabricated lattice filters

V. Conclusion

We propose and fabricate 8 different structures. circuits. Five Mach-Zehnder interferometer structures are used to calculate the group index of the silicon waveguide, two are used as filtering and one is used for calibration. The fabricated MZIs show a good agreement with the simulations result at 1550nm while the fabricated lattice filters show a discrepancy with respect to the simulation. The lattice filters have a non-flat pass band frequency response especially when the FSR is 12nm. This discrepancy could come from the phase response of the directional couplers or path length mismatch of the Mach-Zehnder arms.

References

[1] K. Jinguji and M. Oguma, "Optical half-band filters," in Journal of Lightwave Technology, vol. 18, no. 2, pp. 252-259, Feb. 2000, doi: 10.1109/50.822800.

[2] Zeqin Lu, Han Yun, Yun Wang, Zhitian Chen, Fan Zhang, Nicolas A. F. Jaeger, and Lukas Chrostowski, "Broadband silicon photonic directional coupler using asymmetric-waveguide based phase control," Opt. Express 23, 3795-3808 (2015)